

Fifty-One Publishers, One Election

What it costs to assemble the file no agency publishes, and the half of the country where the answer changes

Democracy's Data

Last updated 31 August 2026

In the week after a presidential election, any news site can tell you the vote in every county in America. It reads like an official national record. It is not, because **no federal agency counts votes**. The Federal Election Commission regulates campaign money. The Election Assistance Commission surveys election administration. Neither is given the returns, and neither publishes them.

What exists instead is fifty-one separate publications by fifty-one separate officers, each certifying their own state under their own law. Every national county table you have ever seen is somebody's private assembly of those publications. So this brief asks a citizen's question rather than a technician's: **where do national election results actually come from — and when two assemblies of the same election sit side by side, do they agree?**

01 The test

The question is answered by doing the assembly. The certified county-level returns of the 2020 and 2024 presidential elections were collected from all fifty-one chief election officers: every state's, plus the District of Columbia's. Beside that assembly sits the file much of the public conversation actually rests on, Tony McGovern's volunteer compilation of county presidential results (https://github.com/tonmcg/US_County_Level_Election_Results_08-24), which several other chapters of this book read too. Comparing the two, county by county, measures what the private assembly step costs.

Certified returns are the right instrument because they are the official record: an officer attested each one under law, which no compilation can say of itself. What a certified return is, who signs it, and what it can never show is the subject of the returns-source chapter, this cluster's data biography.

02 The data

Collecting fifty-one publishers means collecting **102 documents from 51 jurisdictions, across two elections, in 11 distinct file formats**.

Format	Documents
pdf	31

Format	Documents
json	17
csv	15
xlsx	12
html	6
txt	6
xml	6
xls	4
txt/tsv	2
zip/tsv	2
pdf (scanned)	1

32 of the 102 are PDFs, and one of those has **no text layer at all** — it is a photograph of a document, and the only way through it is optical character recognition. Three of the 102 sit on pages that do not say the returns are certified, so they are recorded as `unknown` rather than promoted.

The result is 3,197 reporting units carrying 155,205,623 votes for president, in `pres2024_counties_official.csv` and its 2020 companion. Each row holds `state_name`, `county_fips`, `county_name`, `votes_dem`, `votes_gop` and `total_votes`. Beside the returns sits a record file, `provenance.csv`, one row per document. It holds the address the document came from, the date it was fetched, the format it arrived in, and whether the page claimed certification, with a note on every judgement made while parsing. With fifty-one sources, a citation at the level of the file would be a claim about nothing.

One cleaning decision matters more than the rest. The target was “county-level returns,” and fifty rows refuse the category:

Jurisdiction	Rows	Unit
District of Columbia	8	ward
Maine	1	town or unorganized territory
Missouri	1	city reporting separately from its county
Rhode Island	40	town

Rhode Island publishes by **town**; its five counties have no government and the state does not report to them. The District of Columbia publishes by **ward**. Kansas City reports separately from Jackson County, which contains most of it. None of these is a defect in the source. They are what those places publish, and the assembly keeps them as they are — **with no county code**, because inventing one would be the error rather than the fix.

Before you read on, commit to two numbers. You now have the certified returns from every state, and the same GitHub compilation several other chapters of this book read too, among them `mapping` and `wind-map`. Both describe the 2024 presidential election. **In how many of the roughly 3,100 counties do you expect them to disagree**

about the Democratic vote? And about the total number of votes cast? Write both down.

03 What it says about democracy

Start with the reassuring half. On the major-party vote the two files are in near-perfect agreement:

Quantity	Counties disagreeing	Share
Democratic vote	42	1.3%
Republican vote	47	1.5%
total votes cast	1,588	50.6%

42 counties of 3,138 differ on the Democratic vote and 47 on the Republican. Anybody using either file to say who carried a county will say the same thing almost everywhere.

Then the turn. **1,588 counties report a different number of votes cast** – 50.6% of them, half the country. The two files agree about the numerator and disagree about the denominator: the top number and the bottom number of every rate built from them. That is the more dangerous kind of disagreement, because turnout, margin and share are all computed against the bottom number, and nothing in either file announces the difference.

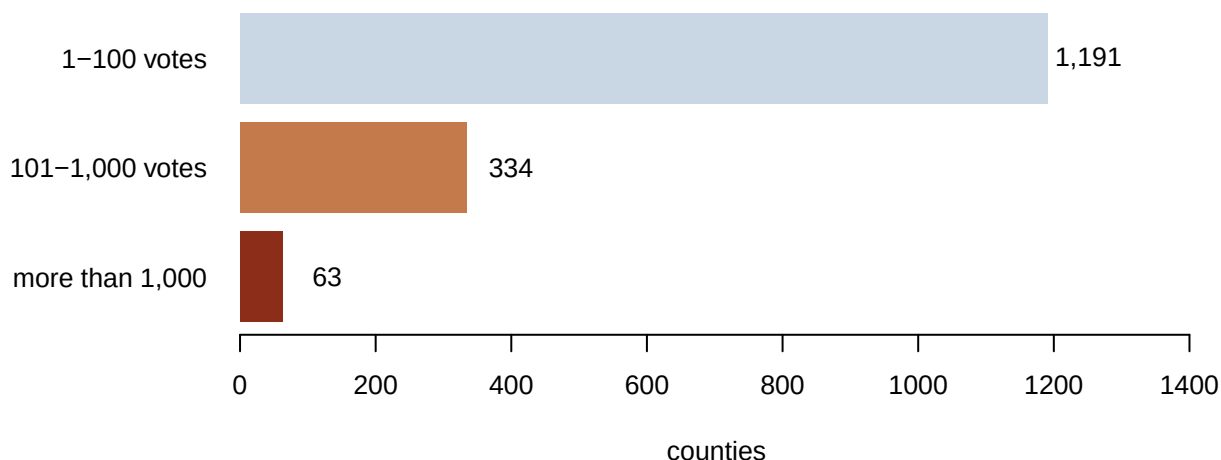


Figure 1. How large the disagreement over votes cast is, in the 1,588 counties where there is one. One bar per size class fits the question, because the question is how the disagreements are distributed – and the answer is that most are a handful of votes and a few are enormous. The median is 21 votes; the gap exceeds a hundred votes in 397 counties and a thousand in 63. Nationally the two totals are 341,884 votes apart.

Three reasons can be named, and they do not add up to all of it.

Dropped third parties. In 107 counties the compilation's total is exactly the Democratic plus the Republican vote, meaning no other candidate is in the denominator. In **102** of those the certified return does record third-party votes. They are not scattered:

State	Counties
New York	61
Alaska	40
Nebraska	1

New York and Alaska between them are nearly all of it – the same two states, and the same defect, that the *returns-source* chapter finds from the other direction. This explains only **6.4%** of the disagreeing counties.

A city that is not in its county. The single largest disagreement is Jackson Missouri:

Source	Total votes
certified state return	193,729
the compilation	317,702

95,660 Democratic votes move depending on which file you opened. Kansas City certifies separately, so the certified assembly keeps it separate and the compilation folds it in. Neither is wrong. Only one of them matches a census population for the same code, and that is the one you need if you are computing turnout.

Write-ins, over-votes and under-votes. What remains is the long tail of small differences, and it is a question of definition rather than of data quality. Is a spoiled ballot, an over-vote or a write-in for a non-candidate a vote cast for president? States answer it differently and a compiler must answer it once.

Worse than a differing number is a differing geography, and Connecticut has one:

Jurisdiction	Certified rows	Compilation rows	Certified unit	Compilation unit
Rhode Island	40	5	Barrington	Bristol County
District of Columbia	8	8	Ward 1	Ward 1
Connecticut	8	9	Fairfield County	Capitol Planning Region

Connecticut appears as 8 counties in one file and 9 planning regions in the other. The state replaced county-equivalents with councils of government for statistical purposes. These are not two codings of one geography; they are different shapes on the ground. A 2020-to-2024 comparison in Connecticut is comparing different objects unless somebody decided otherwise on purpose, and neither file records such a decision.

Rhode Island shows the same seam more quietly: 40 towns in the certified file, 5 counties in the compilation. The compilation is not wrong to aggregate. It is only silent about having done it.

Fips	County	State	Certified	Compilation	Difference
29095	Jackson	Missouri	193,729	317,702	-123,973
17031	Cook	Illinois	2,079,239	2,056,800	22,439

Fips	County	State	Certified	Compilation	Difference
36047	Kings	New York	847,934	835,229	12,705
17043	DuPage	Illinois	460,668	449,660	11,008
36081	Queens	New York	712,191	701,910	10,281
41005	Clackamas	Oregon	246,240	237,953	8,287
53033	King	Washington	1,130,502	1,122,881	7,621
36061	New York	New York	654,372	647,703	6,669
23005	Cumberland	Maine	192,656	199,225	-6,569
36059	Nassau	New York	712,349	706,541	5,808
41051	Multnomah	Oregon	414,250	408,474	5,776
25017	Middlesex	Massachusetts	814,832	809,513	5,319
24031	Montgomery County	Maryland	519,221	513,919	5,302
41017	Deschutes	Oregon	127,375	122,417	4,958
41039	Lane	Oregon	211,285	206,547	4,738

The compilation is careful work by somebody under no obligation to anyone: no correction path, no custodian, no undertaking that it will exist next year. Three chapters of this book read it, and this brief is not an argument against doing so. It is an argument for **saying which file you opened**. “The 2024 vote in Jackson County” is not a fact. “The certified Missouri return for Jackson County, excluding Kansas City” is one, and it differs from the other true statement by 123,973 votes.

04 What this brief cannot tell you

It does not say which file is right, and mostly the question is malformed. Where the two disagree about Kansas City they are answering different questions; where they disagree about a write-in they are applying different definitions. Only the third-party omissions are a defect rather than a choice, and those are 6.4% of the disagreement.

It covers 2024, and 2020 alongside it, and no further back. How much further back this could go is a separate question with an expensive answer. The good long compilations are licensed rather than published, and the public-domain route is the one nobody has assembled, because assembling it is the expensive part.

It does not audit the states against themselves. A certified return is the state's own claim, and this brief takes it at its word. What a canvass can and cannot establish belongs to returns-source.

And the parsing is ours. 32 PDFs were read by script, and a figure recovered from a PDF column is a claim about layout as much as about votes. The state-level crosscheck against an independent wire-service tally is in *derived*/ for exactly that reason: it is the check that would catch a parser that quietly went wrong.

05 What you should have learned

- No federal agency publishes county-level presidential returns. The national record is 51 separate publications in 11 formats, and every national file is a private assembly of them.
- Two assemblies of the same election agree almost everywhere about who won: 42 counties of 3,138 differ on the Democratic vote.
- The same two files disagree about the total votes cast in 1,588 counties — the denominator of every rate — and the gap exceeds a thousand votes in 63 of them.
- The largest single gap, 123,973 votes in Jackson County, Missouri, is a disagreement about the unit rather than the count: one file keeps Kansas City separate and the other folds it in.
- Even the geography can differ: Connecticut is 8 counties in the certified file and 9 planning regions in the compilation, and neither file says so.

06 Extensions

- `largest_gaps.csv` holds the counties where the certified total and the compilation's total differ most, with the gap in `dt`. Take the biggest gap after Jackson County and work out which of the two sources you would trust, and why.
- `third_party_zero.csv` counts, in `counties`, how many of each state's counties report no third-party votes at all. Some states let one candidate appear on several party lines. Is that zero a fact about voters, or a choice about formatting?
- `not_counties.csv` lists, with its `unit` column, the jurisdictions that do not report by county at all. What happens to a national county map if you include them, and what happens if you drop them?
- `formats.csv` counts in `documents` how many publications arrive in each format. Estimate how many hours it would take you to collect all fifty states by hand.
- **Stretch.** Pick one state and find its certified 2016 canvass on the same office's site. How much of the work above would extending the file back one election actually be?
- **Stretch.** Pick one county from `largest_gaps.csv` and find the county's own election office page beneath the state's. Does a third publication of the same number agree with either of the first two?

07 Sources

Data

- The certified returns of all 51 jurisdictions for the 2020 and 2024 presidential elections, published by each state's chief election officer. Every address, retrieval date and format is recorded per jurisdiction-year in this brief's `provenance.csv`; they are too many to list here and that is the brief's point.
- McGovern, Tony. *US County Level Presidential Results 2008–2024*. https://github.com/tonmcg/US_County_Level_Election_Results_08-24
- U.S. Census Bureau. *2024 Gazetteer Files*, counties, for the FIPS resolution and county-name normalisation used when matching state publications to codes. <https://www.census.gov/geographies/reference-files/time-series/geo/gazetteer-files.html>

- Connecticut Office of Policy and Management, on the replacement of counties by councils of government as county-equivalent geography. <https://portal.ct.gov/opm>

The data itself

Every figure above is drawn from these tables. They are plain CSV, they open in a spreadsheet, and they are yours to take further.

`formats.csv`, `largest_gaps.csv`, `not_counties.csv`, `third_party_zero.csv`, `unit_shapes.csv`

REBUILD THIS DATA YOURSELF, WITH AN AI ASSISTANT

I want to rebuild a public dataset from its original sources, without any starting files. Fair warning: this one is a project, not an afternoon. Please do the following and show your work.

THE SOURCE. There is no national file of official county-level presidential returns, because no federal agency counts votes. The returns exist only as fifty-one separate publications – each state's own certified canvass, plus the District of Columbia – and for the 2020 and 2024 elections together that means 102 documents in 11 different formats: spreadsheets, CSVs, JSON feeds, XML, HTML pages, and PDFs. Each state's elections site publishes its own, under its own naming, at its own address. One row of the finished product is one county: its name, its five-digit code, the Democratic and Republican votes, and all votes cast for president.

HOW TO FETCH IT. Send a browser user-agent header; several state sites refuse plain scripted requests without one. One state's site presents a security certificate that does not validate, and the download has to proceed anyway, deliberately. Thirty-two of the documents are PDFs: poppler's `pdftotext`, keeping the layout, turns most of them into parseable text. One – Arizona's signed 2024 canvass – is a scanned image with no text layer at all, so its numbers must come from optical character recognition or by hand. Expect each state to be its own small puzzle.

WHAT TO BUILD.

1. One national table per election: every county with its certified Democratic, Republican, and total presidential vote, stacked from the fifty-one publications.

2. A comparison against the volunteer county compilation on GitHub (tonmcg/US_County_Level_Election_Results_08-24): where the two agree, where they differ, and by how much.

CHECK YOUR WORK. The copies this book used were fetched in August 2026. If your rebuild matches them, you should find:

- 102 documents from 51 jurisdictions, in 11 distinct formats
 - the certified 2024 assembly has 3,197 rows and accounts for 155,205,623 votes for president
 - 3,138 counties can be matched to the compilation; the two disagree on total votes cast in 1,588 of them, the largest single gap is 123,973 votes, and nationally they sit 341,884 votes apart
- States rarely touch a certified canvass after posting it; if a number has moved, a state issued a correction – that is drift in the source, not an error in your work.